1) $\sum_{n=1}^{\infty} \frac{1}{(3n-2)(3n+1)}$

$$a_n = \frac{1}{3}\left(\frac{1}{3n-2} - \frac{1}{3n+1}\right)$$

$$s_n = \sum_{k=1}^{n} a_k = \frac{1}{3}\left(\frac{1}{1} - \frac{1}{4} + \frac{1}{4} - \frac{1}{7} - \frac{1}{7} + \dots + \frac{1}{3n-2} - \frac{1}{3n+1}\right)$$

$$= \frac{1}{3}\left(1 - \frac{1}{3n+1}\right)$$

$$\Rightarrow \lim s_n = \frac{1}{3} \Rightarrow \sum_{n=1}^{\infty} = \frac{1}{3}$$

b) $$\frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right)$$

$$\Rightarrow s_n = \sum_{k=1}^{n} a_k = \frac{1}{2}\left(\frac{1}{1\cdot 2} - \frac{1}{2\cdot 3} + \frac{1}{2\cdot 3} - \frac{1}{3\cdot 4} + \dots + \frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right)$$

$$= \frac{1}{2}\left(\frac{1}{2} - \frac{1}{(n+1)(n+2)}\right) \Rightarrow \lim s_n = \frac{1}{4}$$

$$\Rightarrow \sum_{n=1}^{\infty} = \frac{1}{4}$$

c) $$\sum_{n=1}^{\infty} \frac{1}{n(n+1)\dots(n+p)} = \frac{1}{p}\left(\frac{1}{n\dots(n+p-1)} - \frac{1}{(n+1)\dots(n+p)}\right)$$

$$\Rightarrow s_n = \sum_{k=1}^{n} a_k = \frac{1}{p}\left(\frac{1}{1\cdot 2\dots p} - \frac{1}{2\dots(p+1)} + \dots + \frac{1}{n\dots(n+p-1)} - \frac{1}{(n+1)\dots(n+p)}\right)$$

$$= \frac{1}{p\cdot p!} = 0 \Rightarrow \sum_{n=1}^{\infty} \frac{1}{p\cdot p!}$$

d) $$\sum_{k=1}^{n} \frac{n}{1\cdot 3\cdot 5\dots(2n+1)}$$

$$s_n = \sum_{k=1}^{n} a_k = \sum_{k=1}^{n} \frac{k}{1 \cdot \ldots \cdot (k+1)}$$

$$= \frac{1}{2} \frac{(2k+1)-1}{1 \cdot 3 \cdot \ldots \cdot (2k+1)} = \frac{1}{2}\left(\frac{1}{1 \cdot \ldots \cdot (2k-1)} - \frac{1}{1 \cdot \ldots \cdot (2k+1)}\right)$$

$$= \frac{1}{2}\left(1 - \frac{1}{n+1}\right) = \frac{1}{2}$$

e) $\sum_{n=1}^{\infty} \frac{n \cdot 2^n}{(n+2)!}$

$$s_n = \sum_{k=1}^{n} \frac{n \cdot 2^n}{(n+2)!} = \frac{n \cdot 2^n}{1 \cdot \ldots \cdot (n+2)}$$

$$k+2-1 = k+1$$

$$\frac{(n+2-2) \cdot 2^n}{(n+2)!} = \frac{2^n}{(n+1)!} - \frac{2^{n+1}}{(n+2)!}$$

$$\sum_{k=1}^{n} \frac{2^k}{(k+1)!} - \frac{2^{k+1}}{(k+2)!} = \frac{2}{2!} - \frac{2^2}{3!} + \frac{2^2}{3!} - \ldots$$

$$\frac{2^k}{(k+1)!} - \frac{2^{k+1}}{(k+2)!} \Rightarrow \frac{2}{2!} - \frac{2^{k+1}}{(k+2)!}$$

$$\Rightarrow 1 \qquad \frac{2^{n+2}}{(n+3)!} \cdot \frac{(n+2)!}{2^{n+1}}$$

$$\frac{2}{n+3} = 0$$

f) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)}(\sqrt{n+1}+\sqrt{n})}$

$$\frac{\sqrt{n+1}-\sqrt{n}}{\sqrt{n}\cdot\sqrt{n+1}} = \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}$$

$$1 - \frac{1}{\sqrt{n+1}} = 1$$

g) $\sum_{n=1}^{\infty} \operatorname{arctg} \frac{1}{2n^2}$ $\operatorname{arctg} - \operatorname{arctg} b$

$\operatorname{arct}$

$$\frac{n+1-n}{2n^2} =$$

$\operatorname{arctg} n+1 - \operatorname{arctg}$

$$\frac{2n+1-2n}{4n^2-1} = \frac{1}{4n^2-1}$$

$$\frac{2}{4n^2} = \frac{(2n+1)-(2n-1)}{4n^2-1+1}$$

$\operatorname{arctg}(2n+1) - \operatorname{arctg}(2n-1)$

$= \operatorname{arctg} 3 - \operatorname{arctg} 1 + \ldots + \operatorname{arctg} 2n1$

$- \operatorname{arctg}(2n-1) = -\frac{\pi}{4} + \frac{\pi}{2} = \frac{\pi}{4}$

h) $\sum_{n=1}^{\infty} \operatorname{arctg} \frac{2}{n^2+n+4} = \operatorname{arctg} \frac{2}{n(n+1)+2^2}$

$\operatorname{arctg}$ $\frac{a-b}{ab+1}$

$$n+2-n$$

$$n^2+2n+$$

$$\frac{2}{\frac{n(n+1)}{4}+1} = \frac{\frac{1}{2}}{\frac{n(n+1)}{4}+1}$$

$$\frac{2}{\frac{n(n+1)}{4}+1} = \frac{1}{2}$$

$$\frac{n(n+1)}{4}+1 \qquad n$$

$$n \leq 2 \; ?$$

f) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} =$

$$\sum_{k=1}^{n} 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots + \frac{(-1)^{n+1}}{n}$$

$$a_{2n_1} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots + \frac{1}{2n-1} = \ln 2$$

g) $\sum_{n=1}^{\infty} \frac{1}{(a+n)(a+n+1)+\ldots+(a+n+p)}$

$$\frac{1}{p}\left(\frac{1}{(a+n)!\ldots(a+n+p-1)} - \frac{1}{(a+n+1)\ldots(a+n+p)}\right)$$

$$= \frac{1}{p}\left(\frac{1}{(a+1)!\ldots(a+n+p)} - \frac{1}{(a+2)\ldots(a+n+p+1)} + \frac{1}{\ldots}\right.$$

$$\left.\frac{1}{p}\left(\frac{1}{(a+1)\ldots(a+n+p)} - \frac{1}{(a+n)\ldots}\right)\right.$$

$$= \frac{1}{p(a+1)\ldots(a \quad +n)}$$

h) $\sum_{n=1}^{\infty} \frac{n!}{a(a+1)\dots(a+n-1)}$ 2

$= \rho)\ \operatorname{arctg} \frac{2}{n^2}$ $n+2$

$n^2 - 1 + 1$

$(n-1)(n+1)+1$

$n+1-n+1$

$\frac{n+1-(n-1)}{\cdot n\ n(n+1)-1} = \operatorname{arctg}(n+1) - \operatorname{arctg}(n-1)$

$2 - (1+3-1+4-2+5-3+\dots+(n-1)-(n-3)+n-(n-1)$

$+(n+1) - n - 1$

$\cancel{1} - 1 + n + (n+1)$

$-\frac{\pi}{2} + \cancel{\pi} = \frac{\pi}{2}$

$-\frac{\pi}{4} + \pi = \frac{3\pi}{4}$

m) $\operatorname{arctg} \frac{2^n}{1+2^{2n+1}} = 2$

$\frac{2^{n+1} - 2^n}{1 + 2^{n+1} \cdot 2^n} = 2^n ( \quad = ) \operatorname{arctg} 2^{n+1} - \operatorname{arctg} 2^n$

$4 - 2 + 8 - 4 + \dots$

$2^n - 2^{n+1} + 2^{n+1} - 2^n$

$2^{n+1} - 2$

$= ) \frac{\pi}{2} - \operatorname{arctg} 2$

$$\frac{n^3}{n!} = \frac{n^2}{(n-1)!} = \frac{(n-1)(n+1)+1}{(n-1)!}$$

$$\Sigma \underbrace{\frac{1}{(n-1)!}}_{e} \quad \frac{n^2-1}{(n-1)!}$$

$$\frac{1}{n(2n-1)} \qquad \frac{A}{}(2n-1) + Bn = 1$$

$$A = -1$$
$$B = 2$$

$$\frac{2}{2n-1} - \frac{1}{n} =$$

$$\frac{1}{n-\frac{1}{2}} - \frac{1}{n} =$$

2) a) $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n!)} \cdot a^n, \; a_n = \frac{(n!)^2}{(2n!)} \cdot a^n$

$$\Rightarrow \lim \frac{a_{n+1}}{a_n} = \lim \frac{(n+1)!(n+1)! \cdot a^{n+1}}{(2n+2)!} \cdot \frac{(2n)!}{(n!)(n!) \cdot a^n}$$

$$= \lim \frac{(n+1)^2 \cdot a}{(2n+1)(2n+2)} = \frac{a}{4}, \quad \text{for } a > 4 \Rightarrow \sum_{n=1}^{\infty} a_n \text{ is div}$$

$$a < 4 \Rightarrow \sum_{n=1}^{\infty} a_n \text{ is conv}$$

$$a = 4 \Rightarrow$$

$$\lim n\left(\frac{a_n}{a_{n+1}} - 1\right) = \lim n\left(\frac{4n^2+6n+2}{4n^2+8n+4} - 1\right)$$

$$= \lim n\left(\frac{-2n-4}{4n^2+8n+4}\right) = \lim\left(\frac{-2n^2-4n}{4n^2+8n+4}\right) = -\frac{1}{2} < 1$$

$\Rightarrow \sum_{n=1}^{\infty} a_n$ in conv

b) $\sum_{n=1}^{\infty} \frac{n!}{a(a+1)\dots(a+n)}$, $a_n = \frac{n!}{a(a+1)\dots(a+n)}$

$\Rightarrow \lim \frac{(n+1)!}{a\dots(a+n)(a+n+1)} \cdot \frac{a\dots(a+n)}{n!} = \frac{n+1}{a+n+1} = 1$

$\Rightarrow \lim n\left(\frac{a_n}{a_{n+1}} - 1\right) = n\left(\frac{a+n+1-n-1}{n+1}\right) = \frac{a n}{n+1}$

$= a \Rightarrow \sum_{n=1}^{\infty} a_n$ in conv $a > 1$ and div $a \leq 1$

$a = 1 \Rightarrow \sum_{n=1}^{\infty} \frac{n!}{1 \cdot 2 \cdot \dots \cdot (n+1)} = \frac{1}{n+1}$ div.

c) $\sum_{n=1}^{\infty} \frac{a(a+1)\dots(a+n-1)}{n!} \cdot \frac{1}{n^{\alpha}}$, $a_n = \frac{a\dots(a+n-1)}{n!} \cdot \frac{1}{n^{\alpha}}$

$\lim n\left(\frac{a_n}{a_{n+1}} - 1\right) = \lim n\left(\frac{a\dots(a+n-1)}{n! \cdot n^{\alpha}} \cdot \frac{(n+1)!\,(n+1)^{\alpha}}{a\dots(a+n)} - 1\right)$

$= n\left(\frac{(n+1)\,(n+1)^{\alpha}}{n^{\alpha} \cdot (a+n)} - 1\right)$ $n\left(\frac{(n+1)^{\alpha+1} - n^{\alpha} a - n^{\alpha+1}}{n^{\alpha+1} + a n^{\alpha}}\right)$

$(n+1)^{\alpha+1} = C_{\alpha+1}^{0} n^{\alpha+1} + C_{\alpha+1}^{1} n^{\alpha} \dots$

$= \alpha + 1 - a \Rightarrow \alpha + 1 - a < 1 \Rightarrow \alpha < a$ in div

$\alpha > a$ in conv

d) $\sum_{n=1}^{\infty} \left(\frac{\ln n}{n}\right)^{\alpha} = \frac{\ln^{\alpha} n}{n^{\alpha}}$ ??

e) $\sum_{n=1}^{\infty} \left(\frac{an+b}{cn+d}\right)^{n}$, $a_n = \left(\frac{an+b}{cn+d}\right)^{n}$

$\lim \sqrt[n]{a_n} = \frac{an+b}{cn+d} = \frac{a}{c} \Rightarrow a < c \Rightarrow \sum_{n=1}^{\infty} a_n$ in conv

$a > c \Rightarrow \sum_{n=1}^{\infty} a_n$ in div

$a = c \Rightarrow \lim\left(\frac{an+b-an-d}{an+d}+1\right)^{n\frac{b-d}{an+d}\cdot\frac{an+d}{b-d}} = e^{\frac{b-d}{a}} > 0 \Rightarrow a = c \Rightarrow \sum_{n=1}^{\infty} a_n$ in div

f) $\sum_{n=1}^{\infty} \left(\sqrt{n^2+3n+2}-n\right)^n$, $a_n = \left(\sqrt{n^2+3n+2}-n\right)^n$

$\lim \sqrt[n]{a_n} = \lim \frac{3n+2}{\sqrt{n^2+3n+2}+n} = \frac{3}{2} > 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ in div

g) $\sum_{n=1}^{\infty} \sin \frac{\pi}{n^\alpha}$, $\alpha > 0 \Rightarrow a_n = \sin \frac{\pi}{n^\alpha} \leq \frac{\pi}{n^\alpha} = b_n$

$\Rightarrow \lim \frac{a_n}{b_n} = \frac{\sin \frac{\pi}{n^\alpha}}{\frac{\pi}{n^\alpha}} = 0$

$l = 0 \Rightarrow \alpha > 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ in conv

$0 < \alpha \leq 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ in div.

h) $\sum_{n=2}^{\infty} (2-\sqrt{e})(2-\sqrt[3]{e})\dots(2-\sqrt[n]{e})$, $a_n = (2-\sqrt{e})\dots(2-\sqrt[n]{e})$

$\lim \frac{a_{n+1}}{a_n} = (2-\sqrt[n+1]{e}) = 1$

$\lim n\left(\frac{a_n}{a_{n+1}}-1\right) = n\left(\frac{1}{2-\sqrt[n+1]{e}}-1\right)$

$$= n\left(\frac{1 - 2 + \sqrt[n+1]{e}}{2 - \sqrt[n+1]{e}}\right) = n\left(\frac{\sqrt[n+1]{e} - 1}{2 - \sqrt[n+1]{e}}\right) = \text{??}$$

i) $\sum_{n=1}^{\infty} a^{\ln n}$, $a > 0$, $a_n = a^{\ln n} = n^{\ln a}$

$\Rightarrow a_n = \frac{1}{n^{-\ln a}}$, $a_n$ conv $\Leftrightarrow -\ln a > 1$

$\ln a < -1$

$a < \frac{1}{e}$

$\Rightarrow \sum_{n=1}^{\infty} a_n$ conv $\Leftrightarrow a < \frac{1}{e}$

div $\Leftrightarrow a \geq \frac{1}{e}$

$-\ln a \leq 1$

$\Rightarrow \ln a \geq -1$

$a \geq \frac{1}{e}$

j) $\sum_{n=1}^{\infty} a^{1 + \frac{1}{2} + \ldots + \frac{1}{n}}$, $a > 0$

$a_n = a^{\delta_n + \ln n} = a^{\delta_n} \cdot n^{\ln a}$

k) $\sum_{n=2}^{\infty} (\sqrt[n]{n} - 1)$, $a_n = \sqrt[n]{n} - 1 = e^{\frac{\ln n}{n}} - 1 \geq \frac{\ln n}{n}$

$\geq \frac{1}{n}$ ??

l) $\sum_{n=1}^{\infty} \left(\frac{1 \cdot 3 \cdot \ldots \cdot (2n-1)}{2 \cdot \ldots \cdot 2n}\right)^{\alpha}$, $a_n = \left(\frac{1 \cdot 3 \cdot \ldots \cdot (2n-1)}{2 \cdot 4 \cdot \ldots \cdot 2n}\right)^{\alpha}$

$\lim \frac{a_{n+1}}{a_n} = \left(\frac{2n+1}{2n+2}\right)^{\alpha} = 1$

$\lim n\left(\frac{a_n}{a_{n+1}} - 1\right) = n\left(\frac{(2n+2)^{\alpha} - (2n+1)^{\alpha}}{(2n+1)^{\alpha}}\right)$

$$(2n+2)^\alpha = C_\alpha^0 (2n)^\alpha + C_\alpha^1 (2n)^{\alpha-1} \cdot 2 = \alpha_2^\alpha \cdot n^{\alpha-1}$$

$$(2n+1)^\alpha = C_\alpha^0 (2n^\alpha) + C_\alpha^1 2n^\alpha = \alpha_2^{\alpha-1} \cdot n^{\alpha-1}$$

$$\Rightarrow \lim \frac{\alpha_2^\alpha \cdot n^\alpha - \alpha_2^{\alpha-1} \cdot n^\alpha}{\alpha_2^{\alpha-1} \cdot n^\alpha} = \lim \frac{n^\alpha (\alpha_2^\alpha - \alpha_2^{\alpha-1})}{n^\alpha \; \alpha \cdot 2^{\alpha-1}}$$

$$= \frac{\alpha_2^\alpha - \alpha_2^{\alpha-1}}{\alpha \cdot 2^{\alpha-1}} = \frac{2\alpha - \alpha}{\alpha} = 1 \quad ???$$

m) $\sum_{n=1}^{\infty} \frac{1}{1+\sqrt{2}+\sqrt[3]{3}+\ldots+\sqrt[n]{n}} = \; ??$

n) $\sum_{n=2}^{1} \frac{1}{n \ln n}$ $\qquad a_n = \frac{1}{n \ln n}$, $2^n \cdot a_{2n} = \frac{2^n}{2^n \cdot \ln 2^n}$

$= \frac{1}{n \ln 2}$ which is div $\Rightarrow \sum_{n=1}^{\infty} a_n$ is div

3) $\sum_{n=1}^{\infty} (-1)^{n+1} \cdot \left( \frac{2n+1}{3^n} \right)$